DUMMY TENDER DOCUMENT

MULTIPLE SPLIT-SYSTEM AIR CONDITIONERS AND AIR-TO-AIR HEAT PUMPS

Academic / Project Procurement Simulation

Tender Reference	MSAC-A2AHP/2026/002
Tender Type	Open / Dummy Technical & Commercial Tender
Issue Date	20 September 2026
Bid Validity	90 days from submission deadline
Procuring Entity	ABC HVAC Systems – Academic Simulation
Project	Supply and Installation of Split-System AC & Air-to-Air Heat Pumps
Application	Residential / Light Commercial HVAC

IMPORTANT: This document is a fictional tender prepared for academic/project demonstration only. It is not issued by any government department, PSU, company, or purchasing authority and has no contractual validity.

1. NOTICE INVITING TENDER

ABC HVAC Systems invites sealed technical and commercial offers from eligible manufacturers, authorized distributors and HVAC contractors for the supply, installation, testing and commissioning of multiple split-system air conditioners and air-to-air heat pump systems.

Scope: Complete HVAC systems including outdoor units, indoor units, refrigerant piping, controls, electrical accessories, mounting supports, condensate drainage and commissioning.

2. TENDER INFORMATION

Item	Requirement
Tender Reference	MSAC-A2AHP/2026/002
Equipment	Multiple Split-System ACs and Air-to-Air Heat Pumps
Quantity	20 systems – indicative
Capacity Range	1.5 TR to 3.0 TR equivalent, as scheduled
Delivery Period	6 weeks from purchase order
Installation	Included
Testing & Commissioning	Included
Warranty	Minimum 12 months from commissioning

3. SCOPE OF SUPPLY

• Supply of indoor and outdoor split-system air-conditioning units.

• Supply of air-to-air heat pump systems capable of heating and cooling operation.

• Refrigerant copper piping, insulation, fittings and approved connection accessories.

• Indoor-unit mounting plates/supports and outdoor-unit brackets or floor stands.

• Condensate drain piping and required drainage accessories.

• Electrical interconnections, isolators and control accessories within the defined scope.

• Installation, pressure testing, evacuation, refrigerant charging where required and commissioning.

• Operation manuals, service documentation, warranty certificates and as-installed records.

4. TECHNICAL SPECIFICATION

Parameter	Dummy Tender Requirement
System type	Split-system air conditioner / reversible air-to-air heat pump
Configuration	Indoor unit + outdoor condensing/heat-pump unit
Cooling capacity	Approximately 1.5–3.0 TR per system
Heating capacity	To be stated by bidder; suitable for specified design conditions
Compressor	Variable-speed / inverter type preferred
Refrigerant	R-32 or equivalent approved low-GWP refrigerant
Indoor unit	Wall-mounted or equivalent, with washable filter
Outdoor unit	Air-cooled, weather-resistant construction
Control	Wired/wireless remote control and electronic control system

Parameter	Dummy Tender Requirement
Power supply	230 V AC, 50 Hz, single phase unless otherwise specified
Indoor airflow	Multi-speed fan with automatic airflow control preferred
Temperature control	Electronic thermostat / microprocessor control
Protection	Over-current, high/low pressure and compressor protection as applicable
Energy performance	High-efficiency inverter design preferred
Noise	Low indoor and outdoor sound levels; values to be declared by bidder

5. AIR-TO-AIR HEAT PUMP REQUIREMENTS

• The heat pump shall provide both cooling and heating modes through a reversible refrigeration cycle.

• The system shall use a reversing valve or equivalent arrangement for changeover between heating and cooling.

• Outdoor-unit performance shall be declared at the proposed design ambient conditions.

• Defrost control shall be provided where required by the proposed operating range.

• The supplier shall state rated heating capacity, COP/efficiency data and operating temperature limits.

• Controls shall prevent unsafe compressor operation during mode changes and abnormal operating conditions.

6. INSTALLATION REQUIREMENTS

• Indoor units shall be mounted level and securely fixed to suitable structural surfaces.

• Outdoor units shall be installed with adequate clearance for airflow and service access.

• Refrigerant lines shall be correctly sized, routed and insulated to minimize heat gain/loss.

• All refrigerant joints shall be properly brazed/flared as applicable and leak tested.

• Vacuum evacuation shall be performed before commissioning.

• Condensate drainage shall have adequate slope and shall not cause water leakage.

• Electrical connections shall be properly terminated, protected and earthed.

• The completed installation shall be checked for vibration, abnormal noise and refrigerant leakage.

7. PERFORMANCE & TESTING

Test / Inspection	Acceptance Requirement
Visual inspection	No visible damage, loose parts or installation defects
Dimensional/site inspection	Equipment shall fit the approved installation arrangement
Refrigerant leak test	No leakage at specified test pressure
Vacuum test	System shall achieve and hold the specified evacuation level
Cooling operation	Stable operation at rated/test conditions
Heating operation	Stable heating operation for heat-pump units
Electrical test	Voltage, current, insulation and earthing verified
Controls test	All operating modes and safety controls functional
Condensate test	No leakage or unacceptable water accumulation
Noise/vibration	Within declared/approved limits

8. MATERIAL & CONSTRUCTION REQUIREMENTS

• Indoor and outdoor unit cabinets shall be suitably corrosion-resistant for the intended installation environment.

• Heat exchanger coils shall use materials suitable for the selected refrigerant and operating conditions.

• Insulation shall be continuous and protected against condensation.

• Refrigerant piping shall be refrigeration-grade copper or an approved equivalent.

• Fasteners, brackets and supports shall be appropriate for the equipment mass and installation environment.

• All exposed electrical components shall have appropriate protection against accidental contact.

9. QUALITY ASSURANCE

• Equipment shall be factory tested before dispatch.

• Manufacturer shall maintain serial-number traceability for supplied units.

• Only approved refrigerants and compatible components shall be used.

• Any non-conforming equipment shall be repaired or replaced before final acceptance.

• Installation and commissioning records shall be submitted to the purchaser.

10. DOCUMENTS TO BE SUBMITTED

• Product datasheets and technical catalogue.

• Cooling and heating capacity data.

• Energy-efficiency/performance data.

• General arrangement and installation drawings.

• Refrigerant piping and electrical connection details.

• Pressure/leak and commissioning test reports.

• Operation and maintenance manuals.

• Warranty certificate and service contact details.

11. BOQ / PRICE SCHEDULE

Sr.	Description	Qty.	Unit	Unit Price (■)	Total (■)
1	Split-system air conditioner – complete system	12	Nos.	_____	_____
2	Air-to-air heat pump – complete system	8	Nos.	_____	_____
3	Refrigerant piping & insulation	1	Lot	_____	_____
4	Installation & commissioning	20	Nos.	_____	_____
5	Testing & documentation	1	Lot	_____	_____
	Sub-total				_____
	GST / applicable taxes				_____
	Grand Total				_____

12. COMMERCIAL TERMS

• Quoted prices shall clearly identify equipment, installation, accessories, taxes and transportation.

• The bidder shall state the validity period of the quotation.

• Payment terms: 30% advance, 60% after delivery/inspection and 10% after commissioning, as a dummy academic condition.

• Delivery and installation shall follow the approved project schedule.

• Any exclusions or deviations shall be clearly stated in the commercial offer.

13. WARRANTY & AFTER-SALES SERVICE

The supplier shall provide a minimum 12-month warranty against manufacturing defects from commissioning. The bidder shall state the availability of spare parts, service response procedure and recommended preventive-maintenance schedule. Warranty exclusions shall be clearly identified.

14. BID SUBMISSION CHECKLIST

- Technical offer submitted.
- Equipment datasheets submitted.
- Cooling/heating performance data submitted.
- Installation methodology submitted.
- Testing and commissioning procedure submitted.
- Delivery schedule submitted.
- Commercial price schedule submitted.
- Deviation statement submitted.
- Warranty and after-sales service terms submitted.

15. VENDOR DECLARATION

We hereby declare that the information provided in our technical and commercial offer is accurate to the best of our knowledge and that the proposed equipment can be supplied, installed, tested and commissioned in accordance with the requirements stated in this dummy tender document.

Vendor Name	_____
Authorized Representative	_____
Designation	_____
Signature	_____
Date	_____

END OF DUMMY TENDER DOCUMENT

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